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ITCC503-303i

```
print("Hello World")
```

```
Hello World
```

```
!pip install pandas matplotlib openpyxl
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.13/dist-packages (2.2.3)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (3.10.0)
Requirement already satisfied: openpyxl in /usr/local/lib/python3.13/dist-packages (3.1.5)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.13/dist-packages (from pandas) (2.1.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas) (2026.5)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (4.67.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.5.1)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (3.3.3)
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.13/dist-packages (from openpyxl) (2.0.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

```
import pandas as pd

data = {
    "Month": [
        "January",
        "February",
        "March",
        "April",
        "May",
        "June"
    ],
    "Sales": [15000, 18000, 22000, 25000, 30000, 28000],
    "Expenses": [8000, 9000, 11000, 12000, 15000, 14000]
}

df = pd.DataFrame(data)

df.to_csv("sales_data.csv", index=False)

print("CSV file created successfully!")
```

```
CSV file created successfully!
```

```
df = pd.read_csv("sales_data.csv")

display(df)
```

	Month	Sales	Expenses
0	January	15000	8000
1	February	18000	9000
2	March	22000	11000
3	April	25000	12000
4	May	30000	15000
5	June	28000	14000

```
import pandas as pd
import matplotlib.pyplot as plt

# Read data from CSV file
df = pd.read_csv("sales_data.csv")

# Display the data
print(df)

# Create a line chart
plt.plot(
    df["Month"],
    df["Sales"],
    marker="o",
    label="Sales",
    color="blue",
)

plt.plot(
    df["Month"],
    df["Expenses"],
    marker="s",
    label="Expenses",
    color="red",
)

# Add chart information
plt.title("Monthly Sales and Expenses")
plt.xlabel("Month")
plt.ylabel("Amount ($)")
plt.legend()
plt.grid(True)

# Display chart
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
...       Month  Sales  Expenses
    0   January  15000      8000
    1  February  18000      9000
    2     March  22000     11000
    3     April  25000     12000
    4       May  30000     15000
    5      June  28000     14000
```



```python
import pandas as pd
import matplotlib.pyplot as plt

# URL of online CSV dataset
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/flights.csv"

# Read data directly from the website
df = pd.read_csv(url)

pd.read_csv(url)

# Convert year and month into dates
df["Date"] = pd.to_datetime(
    df["year"].astype(str) + "-" + df["month"],
    format="%Y-%B"
)

# Display first 10 records
print(df.head(10))

# Create line chart
plt.figure(figsize=(12, 6))

plt.plot(
    df["Date"],
    df["passengers"],
    color="blue",
    marker="o",
    markersize=3,
    linewidth=1.5
)

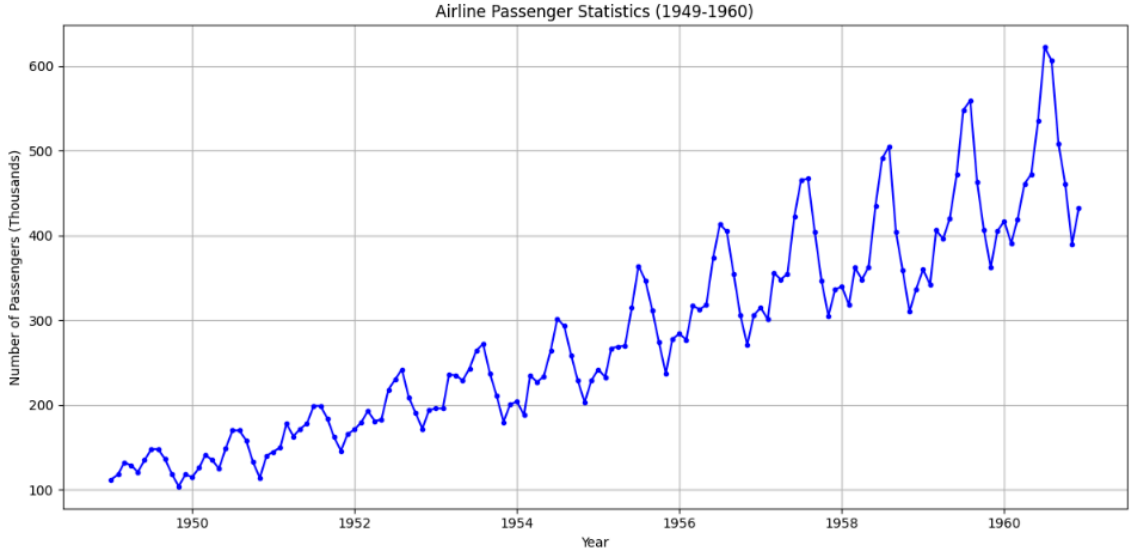
# Chart title and labels
plt.title("Airline Passenger Statistics (1949-1960)")
plt.xlabel("Year")
plt.ylabel("Number of Passengers (Thousands)")

# Add grid
plt.grid(True)

# Improve chart layout
plt.tight_layout()

# Display chart
plt.show()
```

```
    year      month  passengers       Date
0  1949    January         112 1949-01-01
1  1949   February         118 1949-02-01
2  1949      March         132 1949-03-01
3  1949      April         129 1949-04-01
4  1949        May         121 1949-05-01
5  1949       June         135 1949-06-01
6  1949       July         148 1949-07-01
7  1949     August         148 1949-08-01
8  1949  September         136 1949-09-01
9  1949    October         119 1949-10-01
```




```python
import pandas as pd
import matplotlib.pyplot as plt

# Read Excel file
df = pd.read_excel("company_data.xlsx")

# Display records
print(df.head())

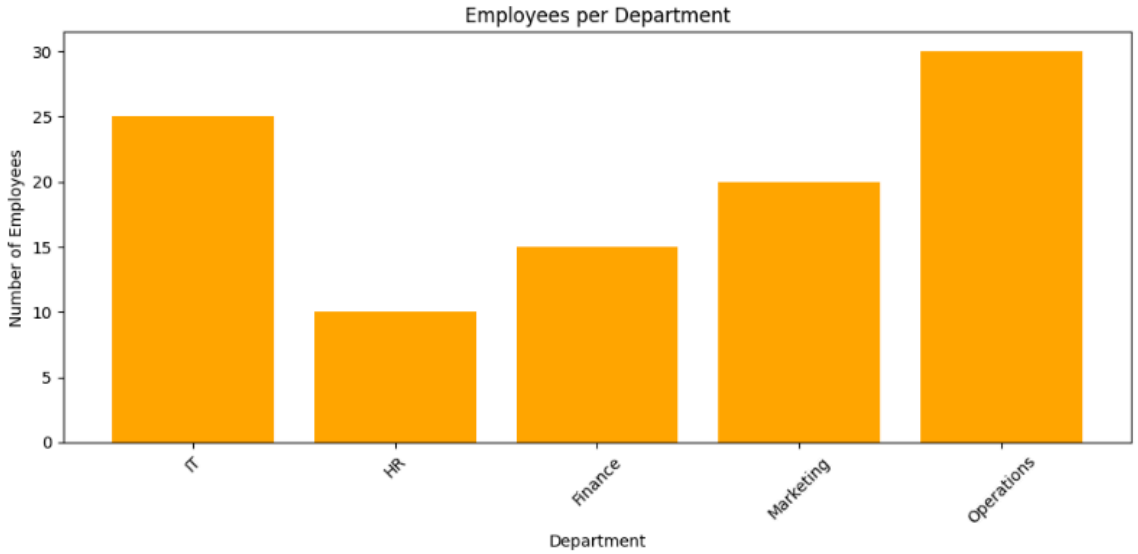
# Create bar chart
plt.figure(figsize=(10, 5))

plt.bar(df["Department"], df["Employees"], color="orange")

plt.title("Employees per Department")
plt.xlabel("Department")
plt.ylabel("Number of Employees")

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
   Department  Employees
0          IT         25
1          HR         10
2     Finance         15
3   Marketing         20
4  Operations         30
```




```
from google.colab import files

uploaded = files.upload()
```

Choose Files company_data.xlsx

company_data.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 8220 bytes, last modified: 10/30/2026 - 100% done

```
Saving company_data.xlsx to company_data (1).xlsx
```